Day 28 – Linked List Problems

🚀 Day 28 – Linked List Problems

Aaj hum ye important problems cover karenge:

- ✓ Remove Nth Node from End
- ✓ Check Palindrome Linked List
- ✓ Merge Two Sorted Linked Lists
- ✓ Intersection Point of Two Linked Lists
- ✓ Remove Duplicates (Sorted LL)

🧠 1. Remove Nth Node from End

💡 Concept (Two Pointer)

👉 Fast pointer ko N steps aage le jao
👉 Phir slow & fast dono ko move karo
👉 Jab fast end pe pahuch jaye → slow ka next delete karo

🔍 Dry Run

List: 1 → 2 → 3 → 4 → 5
n = 2

👉 Remove = 4

💻 Code (Java)

```
1. public Node removeNthFromEnd(Node head, int n) {
    Node dummy = new Node(0);
    dummy.next = head;

    Node fast = dummy;
    Node slow = dummy;

    for (int i = 0; i <= n; i++) {
        fast = fast.next;
    }

    while (fast != null) {
        fast = fast.next;
        slow = slow.next;
    }

    slow.next = slow.next.next;
```

```
    return dummy.next;
}
2.
```

🧠 2. Palindrome Linked List

💡 Concept

👉 Middle find karo
👉 Second half reverse karo
👉 Dono half compare karo

🔍 Example

1 → 2 → 2 → 1 ✅ Palindrome

💻 Code

```
public boolean isPalindrome(Node head) {
    if (head == null || head.next == null) return true;

    Node slow = head, fast = head;

    while (fast != null && fast.next != null) {
        slow = slow.next;
        fast = fast.next.next;
    }

    Node prev = null;
    while (slow != null) {
        Node next = slow.next;
        slow.next = prev;
        prev = slow;
        slow = next;
    }

    Node first = head;
    Node second = prev;

    while (second != null) {
        if (first.data != second.data) return false;
        first = first.next;
        second = second.next;
    }

    return true;
}
```

🧠 3. Merge Two Sorted Linked Lists

💡 Concept

👉 Dono lists compare karo
👉 Chhota element add karo

🔍 Example

List1: 1→3→5
List2: 2→4→6

👉 Result: 1→2→3→4→5→6

💻 Code

```
public Node merge(Node l1, Node l2) {
    Node dummy = new Node(0);
    Node temp = dummy;

    while (l1 != null && l2 != null) {
        if (l1.data < l2.data) {
            temp.next = l1;
            l1 = l1.next;
        } else {
            temp.next = l2;
            l2 = l2.next;
        }
        temp = temp.next;
    }

    if (l1 != null) temp.next = l1;
    if (l2 != null) temp.next = l2;

    return dummy.next;
}
```

🧠 4. Intersection of Two Linked Lists

💡 Concept (Best Trick 🔥)

👉 Dono pointers ko switch karao

👉 A → B
👉 B → A

👉 Equal distance cover karenge → intersection milega

💻 Code

```
public Node getIntersection(Node headA, Node headB) {
    if (headA == null || headB == null) return null;

    Node a = headA;
    Node b = headB;

    while (a != b) {
        a = (a == null) ? headB : a.next;
        b = (b == null) ? headA : b.next;
    }

    return a;
}
```

🧠 5. Remove Duplicates (Sorted LL)

💡 Concept

👉 Same values skip karo

🔍 Example

1 → 1 → 2 → 3 → 3
👉 1 → 2 → 3

💻 Code

```
public Node removeDuplicates(Node head) {
    Node curr = head;

    while (curr != null && curr.next != null) {
        if (curr.data == curr.next.data) {
            curr.next = curr.next.next;
        } else {
            curr = curr.next;
        }
    }

    return head;
}
```